

Stephen Broom

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Software developer focused on building interactive systems and logic-driven applications. Applies strong foundations in C, Java, and Python to create efficient software solutions.

Skills

- **Programming Languages:** C, Java, Python
- **Technical Skills:** Debugging, algorithmic thinking, game logic, state-based systems
- **Tools & Technologies:** Godot, GitHub
- **Computer Science Concepts:** Object-oriented programming, data structures, event-driven programming

Education

Bachelor | University of Central Florida | Orlando
Computer Science | GPA: 3.49

2027 (anticipated)

Projects

Pac-Man Recreation | Turbowarp

April 2025

Developed a full recreation of the classic Pac-Man arcade game using Turbowarp, implementing collision detection, scoring systems, and state-based game progression. Designed and programmed algorithm-driven ghost AI behaviors (chase, scatter, vulnerable modes) and unique player targeting for each of the four enemy ghosts using structured logic and event-driven systems to replicate authentic arcade mechanics. Features a multi-layered UI, including a thematic tutorial system, character selection, and a persistent achievement/shop economy.

Comets (Asteroids Variation) | Java & Swing

Dec 2025

Developed a core game framework in Java, implementing game loop architecture, screen-wrapping logic, and vector-based ship physics including thrust, friction, and rotation. Engineered collision detection systems using Point2D to manage interactions between game entities, utilizing Iterators for efficient object lifecycle management and real-time state updates. Integrated persistent High Score tracking through Java File I/O for data storage and retrieval.

Experience

Administrative Assistant (Part-Time) | Delaney Hotel | Orlando, FL

February 2023

- Managed and maintained digital administrative records using internal software systems
- Coordinated data entry and documentation processes supporting hotel operations

Clinic Assistant (Part-Time) | Florida Spine Care | Orlando, FL

September 2022-May 2023

- Organized and maintained documentation within medical record systems
- Systematically organized physical and digital records to improve office efficiency and facility workflow.